Derivative Securities FNCE 30007

Tutorial 8

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October 22, 2026

Review

Stock pays discrete dividend

- The impact on the stock price: In a perfectly competitive market, stock price is expected to fall on ex-dividend date by dividend amount.
  - Since if you buy the stock afterwards, you will not receive this dividend, therefore, investors require a lower price stock.
  - Let τ denotes ex-dividend date, S_τ as the stock price just after τ, S_{τ^-} as the stock price just before τ, D is the known dividend amount, we have $S_\tau = S_{\tau^-} - D$
- In the imperfect market with transaction cost, tax, etc. The stock price drop will be αD. But in this course, we can assume $\alpha = 1$.

Review

The stock price net of the PV of dividends can be seen as following:

$$S_t^* = \begin{cases} S_t - De^{-r(\tau - t)} & \text{for } t < \tau < T \\ S_t & \text{for } \tau < t < T \end{cases}$$

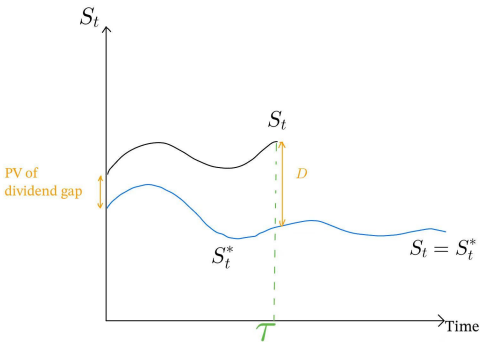


Figure: Stock price path with discrete dividend

Review

The stock price S_t has a jump while the S_t^* doesn't. Therefore, we can treat the stock price as a combination of a risky component S_t^* and a riskless component $PV(div)$ suppose we known the dividend amount and the dividend paying date.

- Therefore, we can model the stock price (the risky part) $S_t^* = S_t - PV(div)$
- My understanding: Note that our previous formula (BSM) is for non-dividend paying stock, in which the whole component is risky. Here, loosely speaking, we should also model the only "risky" part. Since if the stock is only a risk-free bond, then why bother on coming up with BSM to model its price path?

Review

Recall that we have the below formula for the European option with non-dividend paying stock:

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \tag{1}$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \tag{2}$$

where:

$$d_1 = \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{3}$$

$$d_2 = \frac{\log(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} \tag{4}$$

Review

For the dividend paying stock, since we can separate it into the risky position and riskless position, we shall replace the S_0 by the risky position S_0^*.
Note in the discrete dividend payment, $S_0^* = S_0 - PV(D)$

$$c = [S_0 - PV(D)] * N(d_1) - Ke^{-rT}N(d_2) \tag{5}$$

$$p = Ke^{-rT}N(-d_2) - [S_0 - PV(D)] * N(-d_1) \tag{6}$$

where:

$$d_1 = \frac{\log([S_0 - PV(D)]/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{7}$$

$$d_2 = \frac{\log([S_0 - PV(D)]/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} \tag{8}$$

Review

Let's have a look at the formula. Suppose $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$. Suppose the $PV(D)$ varies from $0, 5, 10$, we will see the Call and Put prices as follows:

Table: Call and Put Prices for Varying Dividend Present Value

PV(D)	d_1	d_2	Call Price	Put Price
0.000	0.350	0.150	10.451	5.574
5.000	0.094	–0.106	7.511	7.634
10.000	–0.177	–0.377	5.091	10.214

This makes sense, as the dividend increase, we are more likely to exercise the put option, less lilely to exercise the call option. This push up(down) the price for the put(call).

Review

What about the stock pays continuous dividend? (dividend yield)

- Impact on the stock price:
- Dividend yield q = p.a rate which dividend is paid as pct of stock price.
- Dividend paid discretely: stock price falls at the ex-dividend date.
- Dividend paid continuously: stock price continuously falls at every point (compare with non-dividend paying stock).
  - Equivalently, thinking of the stock price with a growth rate $r - q$
  - r is the risk-free rate since we choose the risk-free rate as the stock price growth rate in the risk-neutral framework.
- Just like we model the stock price's risky part in discrete case, we can also "remove" the riskless part and keep only the risky part. Then the adjusted stock price shall be $S_0 e^{-qT}$

Review

Note in the dividend yield, $S_0^* = S_0 e^{-qT}$

$$c = S_0 e^{-qT} * N(d_1) - K e^{-rT} N(d_2) \tag{9}$$

$$p = K e^{-rT} N(-d_2) - S_0 e^{-qT} * N(-d_1) \tag{10}$$

where:

$$d_1 = \frac{\log(S_0 e^{-qT}/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\log(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{11}$$

$$d_2 = \frac{\log(S_0 e^{-qT}/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\log(S_0/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} \tag{12}$$

Review

Price lower bound (div yield):

- $c \geq S_0 e^{-qT} - Ke^{-rT}$
- $p \geq Ke^{-rT} - S_0 e^{-qT}$

Put-Call Parity (div yield):

- $c + Ke^{-rT} = p + S_0 e^{-qT}$

Tutorial Question

Q1: Consider a stock index currently standing at 250. The dividend yield on the index is 4% per annum, and the risk-free rate is 6% per annum. A three-month European call option on the index with a strike price of 245 is currently worth 10. What is the value of a three-month put option on the index with a strike price of 245, if each index point is valued at $25?

Tutorial Question

In this case $S_0 = 250,\ q = 0.04,\ r = 0.06,\ T = 0.25,\ K = 245$, and $c = 10$. Using put–call parity

$$c + Ke^{-rT} = p + S_0 e^{-qT},$$

or equivalently

$$p = c + Ke^{-rT} - S_0 e^{-qT}.$$

Substituting the numbers,

$$p = 10 + 245\, e^{-0.06\times 0.25} - 250\, e^{-0.04\times 0.25} = 3.84.$$

Multiplying by \$25 per index point,

$$3.84 \times 25 = 96.$$

Tutorial Question

Q2: An index currently stands at 696 and has a volatility of 30% per annum. The risk-free rate of interest is 7% per annum and the index provides a dividend yield of 4% per annum. Calculate the value of a three-month European put with an exercise price of 700, if each index point is valued at $25.

Tutorial Question

In this case $S_0 = 696,\ K = 700,\ r = 0.07,\ \sigma = 0.3,\ T = 0.25$, and $q = 0.04$. We compute

$$d_1 = \frac{\ln(\frac{696}{700}) + (0.07 - 0.04 + \frac{0.3^2}{2})\,0.25}{0.3\sqrt{0.25}} = 0.0868 \approx 0.09,$$

$$d_2 = d_1 - 0.3\sqrt{0.25} = -0.0632 \approx -0.06,$$

and

$$N(-d_1) = 0.4641, \qquad N(-d_2) = 0.5239.$$

The value of the put, p, is then

$$p = 700e^{-0.07\times 0.25} \times 0.5239 - 696e^{-0.04\times 0.25} \times 0.4641 = 40.56,$$

so for \$25 per index point,

$$40.56 \times \$25/\textit{index point} = \$1{,}014.$$

Tutorial Question

Q3: Suppose that the spot price of the Canadian dollar is U.S. $0.95 and that the Canadian dollar/U.S. dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. Calculate the value of a European call option to buy one Canadian dollar for U.S. $0.95 in nine months. Use put-call parity to calculate the price of a European put option to sell one Canadian dollar for U.S. $0.95 in nine months. What is the price of a call option to buy U.S. $0.95 with one Canadian dollar in nine months?

Tutorial Question

In this case $S_0 = 0.95$, $K = 0.95$, $r = 0.05$, $r_f = 0.04$, $\sigma = 0.08$, and $T = 0.75$.

$$d_1 = \frac{\ln(0.95/0.95) + (0.05 - 0.04 + \frac{0.08^2}{2})\,0.75}{0.08\sqrt{0.75}} = 0.1429 \approx 0.14,$$

$$d_2 = d_1 - 0.08\sqrt{0.75} = 0.0736 \approx 0.07,$$

and

$$N(d_1) = 0.5557, \qquad N(d_2) = 0.5279.$$

The Black–Scholes price of the call (with continuous dividend yield r_f) is

$$c = S_0 e^{-r_f T} N(d_1) - K e^{-rT} N(d_2) = 0.95\, e^{-0.04 \cdot 0.75} \times 0.5557 - 0.95\, e^{-0.05 \cdot 0.75} \times 0.5279 = 0.0293$$

i.e. 2.93 cents. By put–call parity,

$$p + S_0 e^{-r_f T} = c + K e^{-rT},$$

so that

$$p = c + K e^{-rT} - S_0 e^{-r_f T} = 0.0293 + 0.95\, e^{-0.05 \cdot 0.75} - 0.95\, e^{-0.04 \cdot 0.75} = 0.0223.$$

The option to buy US$0.95 with C$1.00 is the same as an option to sell one Canadian dollar for US$0.95. This means that it is a put option on the Canadian dollar and its price is US$0.0223.

Tutorial Question

Q4: Suppose that a portfolio is worth \$60 million and the S&P 500 is at 1200. If the value of the portfolio mirrors the value of the index, what options should be purchased to provide protection against the value of the portfolio falling below \$54 million in one year's time, assuming each option contract is for \$100 times the index?

If the value of the portfolio mirrors the value of the index, the index can be expected to have dropped by 10% when the value of the portfolio drops by 10%. Hence when the value of the portfolio drops to \$54 million the value of the index can be expected to be 1080. This indicates that put options with an exercise price of 1080 should be purchased. The options should be on

$$\frac{60\,000\,000}{1200} = \$50{,}000$$

times the index. Each option contract is for \$100 times the index. Hence 500 contracts should be purchased.

Tutorial Question

Q5: Calculate u, d, and p when a binomial tree is constructed to value an option on a foreign currency. The tree step size is one month ($\Delta t = 1/12$), the domestic interest rate is $r = 5\%$ per annum, the foreign interest rate is $q = 8\%$ per annum, and the volatility is $\sigma = 12\%$ per annum.

In this case:

$$a = e^{(r-q)\,\Delta t} = e^{(0.05-0.08)\times\frac{1}{12}} = 0.9975,$$

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.12\sqrt{\frac{1}{12}}} = 1.0352,$$

$$d = \frac{1}{u} = 0.9660,$$

$$p = \frac{a-d}{u-d} = \frac{0.9975-0.9660}{1.0352-0.9660} = 0.4553.$$

Tutorial Question

Q6: Consider an option on a dividend-paying stock when the stock price is \$30, the exercise price is \$29, the risk-free rate is 5% per annum, the volatility is 25% per annum, and the time to maturity is four months. The stock is due to go ex-dividend in 1.5 months and the expected dividend is \$0.50.

- What is the price of the option if it is a European call?
- What is the price of the option if it is a European put?
- What happens to the price if the stock is expected to pay another 50 cents dividend in 6 months (ex-dividend date in 6 months)?

Tutorial Question

European call.
First subtract the present value of the dividend:

$$S_0' = 30 - 0.5\, e^{-0.05\times(1.5/12)} = 29.5031.$$

With $T = 4/12 \approx 0.3333$, define

$$d_1 = \frac{\ln(\frac{S_0'}{K}) + (r + \frac{1}{2}\sigma^2)\, T}{\sigma\sqrt{T}} = \frac{\ln(29.5031/29) + (0.05 + 0.25^2/2)\, 0.3333}{0.25\sqrt{0.3333}} \approx 0.31,$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.3068 - 0.25\sqrt{0.3333} = 0.1625 \approx 0.16.$$

From the standard normal CDF,

$$N(d_1) = 0.6217, \qquad N(d_2) = 0.5636.$$

Hence the call price is

$$c = S_0' N(d_1) - Ke^{-rT} N(d_2) = 29.5031\times 0.6217 - 29\, e^{-0.05\times 0.3333}\times 0.5636 = 2.27.$$

Tutorial Question

European put.
Using put–call parity or directly

$$N(-d_1) = 0.3783, \qquad N(-d_2) = 0.4364,$$

the put price is

$$p = Ke^{-rT} N(-d_2) - S_0' N(-d_1) = 29\, e^{-0.05\times 0.3333}\times 0.4364 - 29.5031\times 0.3783 = 1.28.$$

The answer would stay the same as the next dividend will be paid after the option is expired.